WORKED SOLUTION

STEP III 2000, Question 4

Pure • Rational Function Sketching

STEP III 2000, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/00-s3-q4

The key idea

Setting $f(x) = x + c$ and clearing the denominator, the cubic terms cancel and we are left with a *quadratic* in x. Whether the line meets the curve then comes down to the sign of a discriminant, and that discriminant is itself a quadratic in c, so we take a discriminant of a discriminant. For the sketches, splitting f into a linear part plus partial fractions tells us everything: the asymptotes, which side of them the curve lies, and where it crosses the oblique one.

Solution

Where the line meets the curve

For $x \neq \pm 1$, the equation $f(x) = x + c$ is equivalent to

$$x(x-2)(x-a) = (x+c)(x^2-1).$$

Expanding, $x^3 - (a+2)x^2 + 2ax = x^3 + cx^2 - x - c$. The x^3 terms cancel, leaving

$$q(x) \equiv (a+2+c)x^2 - (2a+1)x - c = 0. \qquad (*)$$

So the line meets the curve exactly when $(*)$ has a real root other than $x = \pm 1$.

? **Ask yourself** Could a root of $(*)$ be one of the excluded values $x = \pm 1$?

Substituting, $q(1) = (a+2+c) - (2a+1) - c = 1 - a$ and $q(-1) = (a+2+c) + (2a+1) - c = 3(a+1)$. Neither depends on c, so $x = 1$ is a root only when $a = 1$, and $x = -1$ is a root only when $a = -1$. For $|a| \neq 1$ every real root of $(*)$ gives a genuine intersection.

The case $a + 2 + c = 0$. Then $(*)$ is linear: $-(2a+1)x + (a+2) = 0$. If $|a| > 1$ then $2a + 1 \neq 0$, so there is the root $x = \frac{a+2}{2a+1}$.

The case $a+2+c \neq 0$. Then $(*)$ is a genuine quadratic with discriminant

$$D(c) = (2a+1)^2 + 4c(a+2+c) = 4c^2 + 4(a+2)c + (2a+1)^2.$$

Think of D as a quadratic in c with positive leading coefficient. Its own discriminant is

$$\begin{aligned} 16(a+2)^2 - 16(2a+1)^2 &= 16[(a+2)-(2a+1)][(a+2)+(2a+1)] \\ &= 16(1-a)\cdot 3(1+a) = 48(1-a^2). \end{aligned}$$

If $|a| > 1$, this is negative, so $D(c) > 0$ for every c; then $(*)$ has two distinct real roots, neither equal to ± 1. Together with the linear case, the line $y = x + c$ meets the curve for every c.

If $|a| = 1$, $D(c) \geq 0$ for every c, so $(*)$ still always has real roots, which is the algebraic content of the claim. But here the curve itself degenerates: a factor cancels and the curve is a hyperbola with a single point missing. For instance, with $a = 1$ we get $f(x) = x - 3 + \frac{3}{x+1}$ $(x \neq 1)$, and the line $y = x - 3$ (its asymptote) does not meet it; in $(*)$ this is $c = -3$, where the only root is the excluded $x = 1$. So for $|a| = 1$ the statement holds for all but these exceptional values of c, and it holds without exception for $|a| > 1$.

If $|a| < 1$, the quadratic $D(c)$ has two real roots and is negative strictly between them. An explicit choice is $c = -\frac{1}{2}(a+2)$, the midpoint of the roots:

$$D(-\tfrac{1}{2}(a+2)) = (a+2)^2 - 2(a+2)^2 + (2a+1)^2 = 3(a^2-1) < 0.$$

For this c, $a + 2 + c = \frac{1}{2}(a+2) \neq 0$ (since $a > -1$), so $(*)$ is a genuine quadratic with negative discriminant and has no real roots. Hence the line $y = x - \frac{1}{2}(a+2)$ misses the curve, as required.

The oblique asymptote

Dividing $x^3 - (a+2)x^2 + 2ax$ by $x^2 - 1$,

$$f(x) = x - (a+2) + \frac{(2a+1)x - (a+2)}{x^2 - 1}.$$

The fraction tends to 0 as $x \to \pm\infty$, so the oblique asymptote is

$$y = x - a - 2.$$

For the sketches it helps to split the fraction into partial fractions:

$$f(x) = x - (a+2) + \frac{a-1}{2(x-1)} + \frac{3(a+1)}{2(x+1)}.$$

This gives us everything:

- zeros at $x = 0,\ 2,\ a$, and vertical asymptotes at $x = \pm 1$;
- the curve meets its oblique asymptote only where $(2a+1)x = a+2$, that is at $x_0 = \frac{a+2}{2a+1}$ (this is the linear case of $(*)$ again);

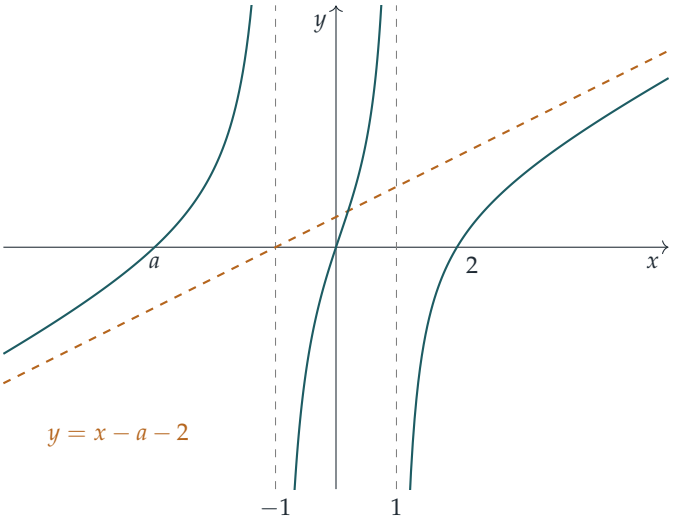
- for large $|x|$ the gap $f(x) - (x - a - 2)$ behaves like $\frac{2a+1}{x}$. In both cases $2a + 1 < 0$, so the curve lies *below* the asymptote as $x \to +\infty$ and *above* it as $x \to -\infty$;
- the coefficients $\frac{a-1}{2}$ and $\frac{3(a+1)}{2}$ fix the behaviour on each side of $x = 1$ and $x = -1$.

Case (i): $a < -1$

Both coefficients are negative: $\frac{a-1}{2} < 0$ and $\frac{3(a+1)}{2} < 0$. So $f(x) \to +\infty$ as $x \to 1^-$ or $x \to -1^-$, and $f(x) \to -\infty$ as $x \to 1^+$ or $x \to -1^+$. Moreover

$$f'(x) = 1 - \frac{a-1}{2(x-1)^2} - \frac{3(a+1)}{2(x+1)^2} > 0,$$

since both subtracted terms are negative, so f is increasing on each of its three branches and there are no turning points at all. The crossing point satisfies $x_0 + 1 = \frac{3(a+1)}{2a+1} > 0$ and $x_0 - 1 = \frac{1-a}{2a+1} < 0$, so $-1 < x_0 < 1$: the curve crosses its asymptote on the middle branch. The zeros are at $a < -1$ (left branch), 0 (middle) and 2 (right).



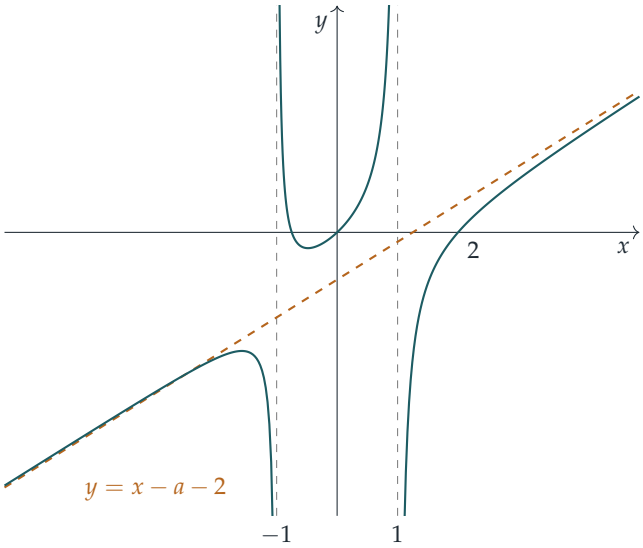
(Drawn for $a = -3$, where the asymptote is $y = x + 1$ and $x_0 = \frac{1}{5}$.)

Case (ii): $-1 < a < -\frac{1}{2}$

Now $\frac{a-1}{2} < 0$ but $\frac{3(a+1)}{2} > 0$. Near $x = 1$ the behaviour is as before ($+\infty$ from the left, $-\infty$ from the right), but near $x = -1$ it flips: $f(x) \to -\infty$ as $x \to -1^-$ and $f(x) \to +\infty$ as $x \to -1^+$. Now $x_0 + 1 = \frac{3(a+1)}{2a+1} < 0$, so $x_0 < -1$: the crossing is on the left branch. The zeros a and 0 both lie in $(-1, 0)$.

Ask yourself Where must the turning points be, even though we are not asked to find them?

On the left branch, the curve comes up from $-\infty$ above the asymptote, crosses it at x_0, then falls to $-\infty$ as $x \to -1^-$; so it has a maximum. It has no zeros there (all three zeros are > -1), so the whole branch lies below the x-axis. On the middle branch $f \to +\infty$ at both ends, it dips below the axis between a and 0 and is positive elsewhere, so there is a minimum between a and 0. On the right branch, $\frac{3(a+1)}{2(x+1)^2} < \frac{3}{4} \cdot \frac{1}{4} < 1$ for $x > 1$, so $f'(x) > 0$: the curve rises from $-\infty$ through $x = 2$ and approaches the asymptote from below.



(Drawn for $a = -\frac{3}{4}$: asymptote $y = x - \frac{5}{4}$, crossing at $x_0 = -\frac{5}{2}$, and a shallow dip below the axis between $-\frac{3}{4}$ and 0.)

Checks

- With $a = 0$ and $c = -1$ (our choice $c = -\frac{1}{2}(a+2)$), $(*)$ becomes $x^2 - x + 1 = 0$, whose discriminant is -3: no intersection, as predicted.
- With $a = -3$: $f(-1.1) \approx 30.9$ and $f(-0.9) \approx -28.8$, which matches the signs found for $x \to -1^{\mp}$ in case (i). With $a = -\frac{3}{4}$, $f(-1.1) \approx -5.7$ and $f(-0.9) \approx 2.1$, the flipped behaviour of case (ii).
- Counting sign changes of f' numerically for several a in each range confirms no turning points in case (i), and exactly one on each of the left and middle branches in case (ii).

Common slips

- Forgetting that roots $x = \pm 1$ of $(*)$ are not intersections; this is exactly what goes wrong when $|a| = 1$.

- Ignoring the case $a + 2 + c = 0$, where $(*)$ is linear and the discriminant argument does not apply.
- Getting the side of the vertical asymptote $x = -1$ wrong in case (ii): the sign of $a + 1$ changes the picture there.

Beyond STEP

When $|a| < 1$, the two roots of $D(c)$ give lines $y = x + c$ that are tangent to the curve, and the lines between them miss it. As a crosses ± 1 these tangents appear or disappear and the curve changes shape: a zero passes through a vertical asymptote and the behaviour beside it flips. Studying how the qualitative shape of a family of curves changes with a parameter is the start of bifurcation theory, met in first-year courses on differential equations; see Boyce and DiPrima, *Elementary Differential Equations*. For more practice at this kind of sketching, Siklos, *Advanced Problems in Mathematics*, is excellent.

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